Predicting U.S. Market Returns Using Fundamental Indicators

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AlgoGators Capstone Project

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	coef	std err	t	P>|t|	[0.025	0.975]
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X2	-4.147e-06	6.57e-07	-6.312	0.000	-5.46e-06	-2.83e-06
X3	8.431e-05	4.15e-06	20.293	0.000	7.6e-05	9.26e-05

1 Abstract

This project investigates the predictability of U.S. stock market returns using a linear regression model based on fundamental economic indicators. The model includes household financial assets, inflation rate, corporate earnings, the cyclically adjusted price-to-earnings (CAPE) ratio, and the risk-free rate. Using annual data from 1947-2024 sourced from the Fed, we estimate the market return as a function of these macro-financial variables. Results suggest that financial asset levels and inflation are statistically significant predictors of future market returns, while earnings and CAPE provide additional explanatory power. The regression model achieves an R-squared of over 0.9 which is highly suspicious, indicating strong in-sample fit, though serial correlation and multicollinearity may limit out-of-sample robustness. This study provides a quantitative foundation for using publicly available macro indicators to model equity market behavior.

2 Introduction

Forecasting equity market returns remains a central question in asset allocation and macro-finance. Traditional asset pricing theories suggest that market returns are influenced by risk premia, economic fundamentals, and investor expectations. Numerous studies have identified predictors including dividend-price ratios, earnings yields, interest rates, and macroeconomic variables (Campbell & Shiller, 1988; Fama & French, 1989). This project adopts a data-driven approach to model annual U.S. market returns using five variables: household financial assets, inflation, earnings, CAPE ratio, and the risk-free rate.

The hypothesis is that aggregate household financial positioning (as a proxy for investor risk tolerance and wealth), macroeconomic price stability (via inflation), corporate earnings, valuation (via CAPE), and interest rates (cost of capital) should together help explain market return fluctuations.

To build further intuition, consider the investor decision cycle: each investor, upon observing macro and market signals, forms an estimate of the asset's fair value. Given the heterogeneity of beliefs, the collective investor base generates a distribution of estimated prices, denoted by f(p). Based on their individual valuation relative to the market price, investors decide whether to buy or sell. This buying and selling behavior,

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aggregated across investors, ultimately shifts the market price, completing a dynamic cycle of expectation and action. Assuming buyers and sellers are drawn from the same group, the equilibrium price p∗ occurs when capital-weighted demand equals supply, captured by:

$$\text{Seller's capital} \cdot \int_0^{p^*} f(p)\,dp = \text{Buyer's capital} \cdot \int_{p^*}^{\infty} f(p)\,dp$$

Following the definition of f(p), $\int_{p^*}^{\infty} f(p)\,dp$ can be interpreted as the **probability of purchase of buyers**. To simplify the notation, we replace $\int_{p^*}^{\infty} f(p)\,dp$ with Pr(buy). Then, rearranging the equation, the market value can be expressed as:

$$\text{Market Value} = (\text{Seller's Capital} + \text{Buyer's Capital}) \cdot \Pr(\text{buy})$$

Applying this to the U.S. equity market—specifically the S&P 500—if we assume that U.S. households are the primary participants (an approximation supported by their ~80% ownership share), then total available capital can be approximated by **household financial assets** (as reported by the Federal Reserve). Assuming further that the S&P 500 index is proportional to the total equity market, we arrive at a simplified relation:

$$\text{Household Equity Holdings} = \Pr(\text{buy}) \cdot \text{Household Financial Assets}$$

$$\text{S\&P } 500_t \propto \text{Household Equity Holdings}_t$$

$$\text{S\&P } 500 \propto \Pr(\text{buy}) \cdot \text{Household Financial Assets}$$

This formulation reduces the forecasting problem to predicting two key components: (1) **household financial asset levels**, and (2) the **probability of purchase**, which may be driven by macroeconomic conditions, valuations, or investor sentiment. This cycle-based perspective offers a theoretical foundation for why financial capital and sentiment-sensitive variables are central to return prediction.

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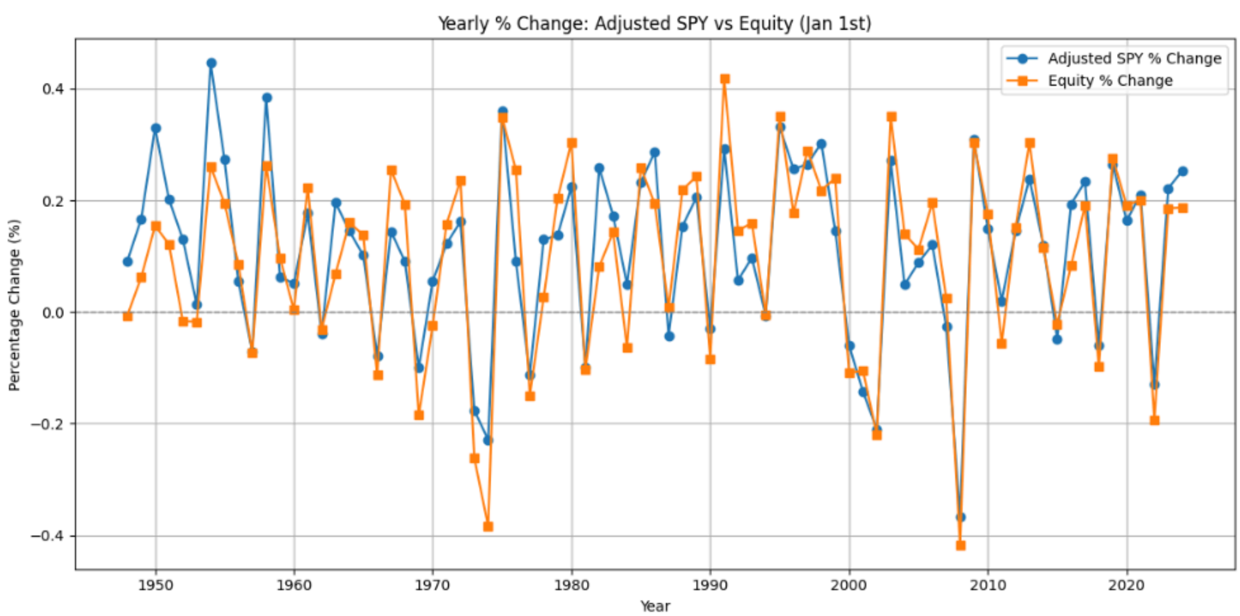
To validate this theoretical framework, we examine whether the year-over-year percentage change in U.S. household equity holdings aligns with the change in the S&P 500 index. From the assumptions, it follows that the growth rates of household equity and the S&P 500 should exhibit similar trends. The chart below plots the annual percentage change of adjusted S&P 500 values and household equity holdings (measured on January 1st each year). The two series demonstrate strong co-movement across decades, suggesting that changes in household equity—driven by both capital availability and investor sentiment—are closely reflected in the S&P 500 index. This visual consistency supports the core assumption of our model: that the market index can be reasonably modeled as a function of household capital and behavioral purchase probability.



3 Methodology

Building on the earlier framework, we assume that **the probability of purchase** is a function of macroeconomic and valuation signals, specifically:

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$$\Pr(\text{buy})_t = \beta_1(\text{CAPE Yield}_t - \text{Risk-Free Rate}_t) + \beta_2 \cdot \text{Inflation}_t + \beta_0$$

This reflects that investors are more likely to allocate capital to equities when valuation is attractive (high CAPE yield relative to the risk-free rate) and inflation is stable or low.

We also assume that **household financial assets** grow at a steady rate over time, such that:

$$\text{Financial Asset}_t = \beta_3 \cdot \text{Financial Asset}_{t-1}$$

Substituting these into the expression for S&P 500 market value:

$$\text{S\&P 500}_t = \Pr(\text{buy})_t \cdot \text{Financial Asset}_t$$

Expanding:

$$\text{S\&P 500}_t = [\beta_1(\text{CAPE Yield}_t - \text{Risk-Free Rate}_t) + \beta_2 \cdot \text{Inflation}_t + \beta_0] \cdot \text{Financial Asset}_t$$

To model **returns**, we normalize by the previous year's S&P 500 value:

$$\text{Return}_t = \frac{\text{S\&P 500}_t}{\text{S\&P 500}_{t-1}} - 1$$

$$\Rightarrow \text{Return}_t \propto [\beta_1(\text{CAPE Yield}_t - \text{Risk-Free Rate}_t) + \beta_2 \cdot \text{Inflation}_t + \beta_0] \cdot \frac{\text{Financial Asset}_t}{\text{S\&P 500}_{t-1}}$$

This final expression yields a regression-ready form for predicting market returns using macroeconomic signals and capital availability:

$$\text{Return}_t = (\beta_1(\text{CAPE Yield}_t - \text{Risk-Free Rate}_t) + \beta_2 \cdot \text{Inflation}_t + \beta_0) \cdot \frac{\text{Financial Asset}_t}{\text{S\&P 500}_{t-1}} + \varepsilon_t$$

To empirically estimate the model, we use annual data from 1947 to 2024 obtained from the Federal Reserve (FRED), which includes U.S. household financial assets, the inflation rate, CAPE ratio, corporate earnings, and the 1-year Treasury yield (risk-free rate). By expanding the previous expression, we obtain a linear regression model of the form:

$$\text{Return}_t = \gamma_1 \cdot \frac{\text{Financial Asset}_t \cdot (\text{CAPE Yield}_t - \text{Risk-Free Rate}_t)}{\text{S\&P 500}_{t-1}} + \gamma_2 \cdot \frac{\text{Financial Asset}_t \cdot \text{Inflation}_t}{\text{S\&P 500}_{t-1}} + \gamma_0 \cdot \frac{\text{Financial Asset}_t}{\text{S\&P 500}_{t-1}} + \varepsilon_t$$

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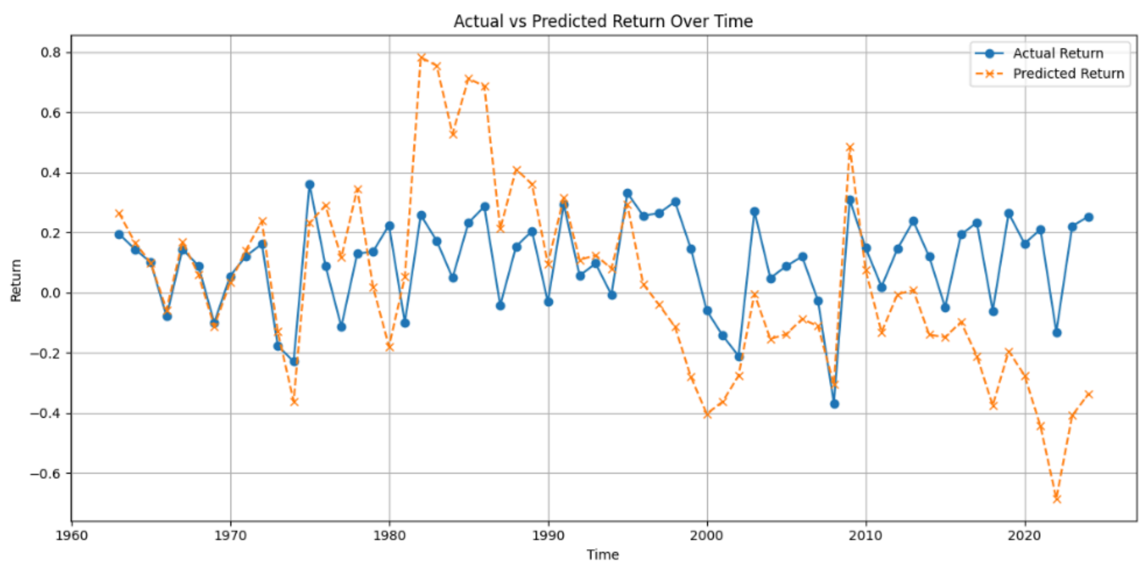
This form allows for a straightforward implementation using **Ordinary Least Squares (OLS)** regression. The resulting model coefficients are shown in the table below, and the fitted values are compared to actual S&P 500 returns over time. As seen in the plot, the model captures broad trends in market returns, especially during major turning points, although it slightly overestimates volatility in certain decades.

4 Results



The regression results derived from the proposed predictive model are summarized in the table above. The three explanatory variables correspond to the terms:

- X_1: $\frac{\text{Financial Asset}_t \cdot (\text{CAPE Yield}_t - \text{Risk-Free Rate}_t)}{\text{S\&P } 500_{t-1}}$
- X_2: $\frac{\text{Financial Asset}_t \cdot \text{Inflation}_t}{\text{S\&P } 500_{t-1}}$
- X_3: $\frac{\text{Financial Asset}_t}{\text{S\&P } 500_{t-1}}$

The regression output shows that X2(inflation interaction term) has a **significantly negative** coefficient, suggesting that inflation erodes expected market returns, consistent with macroeconomic theory. X3, representing normalized financial capital, is

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highly significant and **positively associated** with returns, indicating that increases in household financial asset levels contribute strongly to return expectations. The CAPE yield differential term (X1) is statistically insignificant, suggesting that valuation alone may not be sufficient for short-term prediction in this linear setting.

The accompanying plot compares actual vs. predicted returns over time. The model captures general trends and turning points in the market—particularly during periods of high volatility such as the early 1980s and post-2008. However, the predicted returns tend to **overestimate the magnitude of fluctuations**, especially in more recent years, suggesting potential **model overfitting or omitted variable bias**. The gap in recent decades may also reflect changing market dynamics, such as increased institutional participation, globalization, or nonlinear effects not captured in the current model.

Overall, while the model demonstrates good explanatory power and statistically significant results for key macroeconomic terms, refinements may be needed to improve robustness and out-of-sample forecasting accuracy.

5 Discussion

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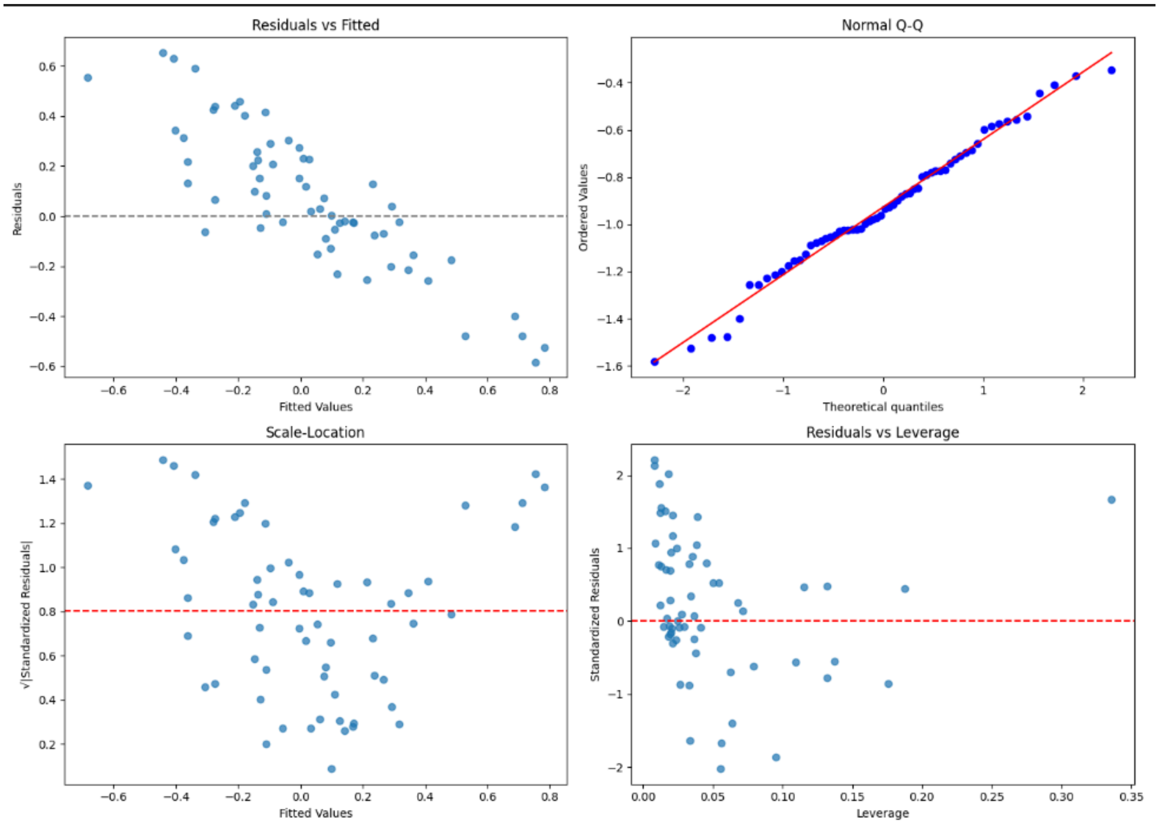

To evaluate the assumptions underlying the linear regression model, we examine standard diagnostic plots. The **Residuals vs Fitted** plot shows a downward-sloping funnel pattern, suggesting potential **non-linearity** and **heteroskedasticity** in the residuals. This violates the assumption of constant variance and may imply that the model performs differently across varying levels of predicted return.

The **Normal Q-Q plot** indicates that residuals are approximately normally distributed, particularly around the center. However, deviations in the tails imply mild **non-normality**, which may affect inference validity for extreme return scenarios.

The **Scale-Location plot** (also known as the Spread-Location plot) further confirms **non-constant variance**, as the spread of standardized residuals appears to increase

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slightly with fitted values. This heteroskedasticity suggests that the error variance is not uniform across the prediction range, possibly reducing the efficiency of OLS estimates.

Lastly, the **Residuals vs Leverage plot** identifies no extreme outliers or high-leverage points with disproportionate influence. Most observations lie within a reasonable leverage range, indicating that no single point dominates the regression fit.

Overall, while the regression performs reasonably well in fitting the central trend, the presence of heteroskedasticity and some non-linearity suggests that a more robust or nonlinear model may improve predictive accuracy and inference stability in future work.

6 Conclusion

This study presents a theoretically grounded and empirically tested model for forecasting U.S. equity market returns using macro-financial variables. By linking the S&P 500 index to household financial capital and investor sentiment—proxied through CAPE yield, inflation, and the risk-free rate—the model captures the fundamental dynamics of price formation under aggregate investor behavior.

The regression analysis demonstrates statistically significant relationships, especially for household capital and inflation effects, and the model performs reasonably well in tracking historical return patterns. However, diagnostic tests reveal signs of heteroskedasticity and non-linearity, suggesting room for improvement in model specification.

Going forward, the primary objective is to enhance the model's **predictive accuracy** through methods such as regularization, nonlinear transformations, or machine learning techniques. Only after the model demonstrates consistent and reliable forecasting power—particularly in out-of-sample performance—can it be confidently adapted into a **practical investment strategy**. Such a strategy could involve dynamic asset allocation or macro signal-driven portfolio construction, offering a disciplined, data-informed approach to navigating equity markets.

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7 References

Campbell, J. Y., & Shiller, R. J. (1988). The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors. Review of Financial Studies, 1(3), 195–228.

Fama, E. F., & French, K. R. (1989). Business Conditions and Expected Returns on Stocks and Bonds. Journal of Financial Economics, 25(1), 23–49.

Shiller, R. J. (2020). Online Data: http://www.econ.yale.edu/~shiller/data.htm

Federal Reserve Bank of St. Louis (FRED). https://fred.stlouisfed.org

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